

ServiceNow Integration Manual

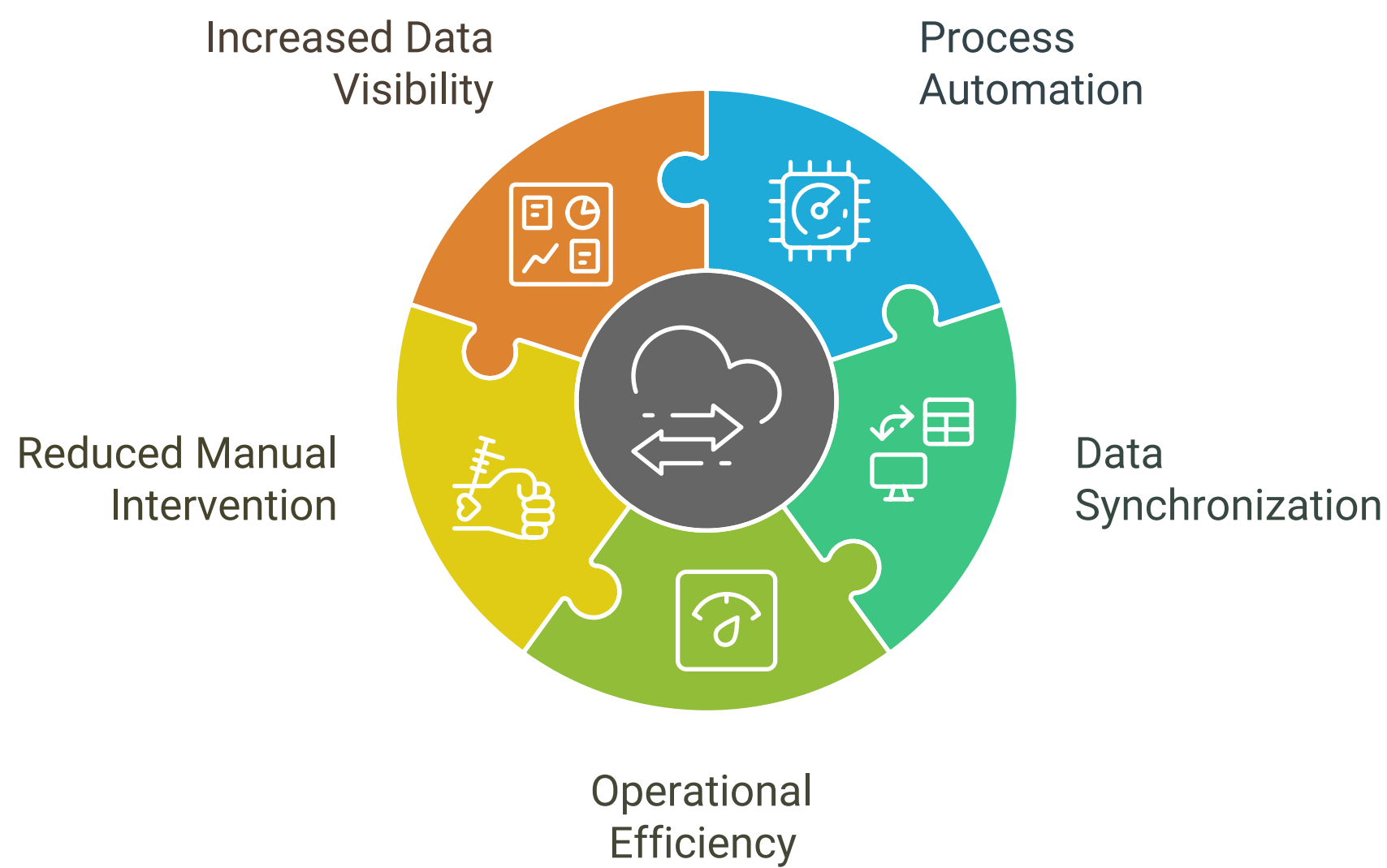
1. Introduction

Purpose of Integration

Integrating ServiceNow with external systems is a critical step for organizations that want to automate processes, synchronize data, and ensure a seamless flow of information.

Integrations improve operational efficiency, reduce manual intervention, and increase the visibility of data across various business applications.

Benefits of ServiceNow Integration



Overview of Integration in ServiceNow

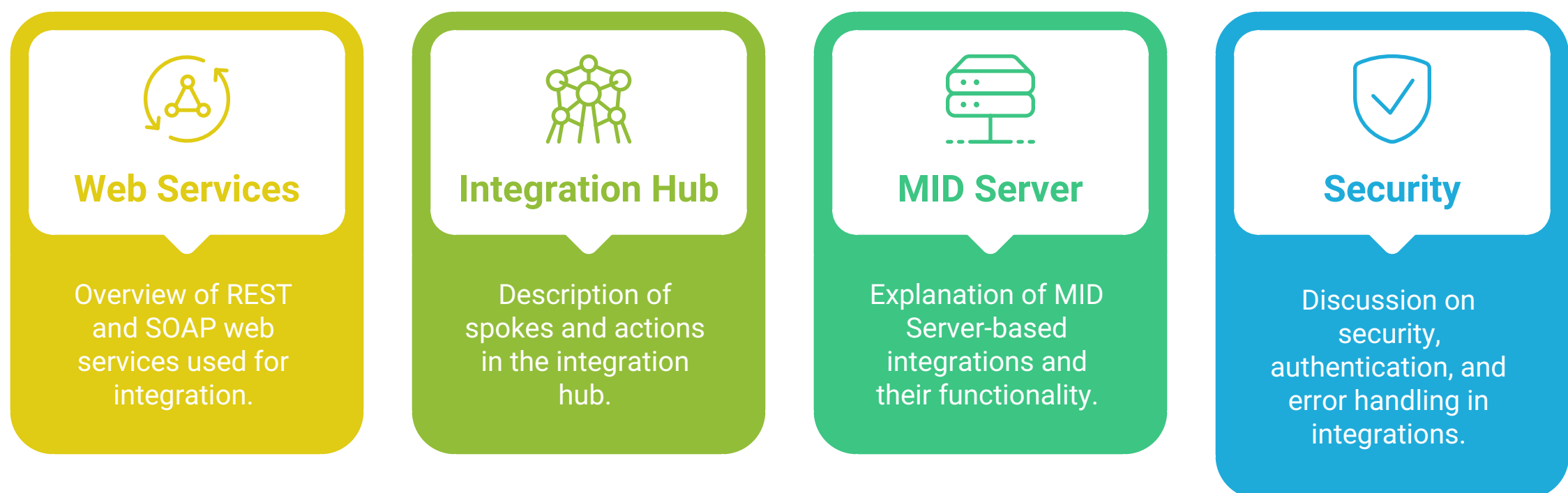
ServiceNow provides several methods for integrating with external systems, including REST and SOAP APIs, Integration Hub spokes, and MID Servers. By using these methods, ServiceNow can interact with other systems such as databases, third-party ITSM tools, HR systems, cloud platforms, and security solutions.

Scope

This manual covers various integration methods and use cases, including:

- REST and SOAP Web Services
- Integration Hub (Spokes and Actions)
- MID Server-based integrations
- Security, authentication, and error handling

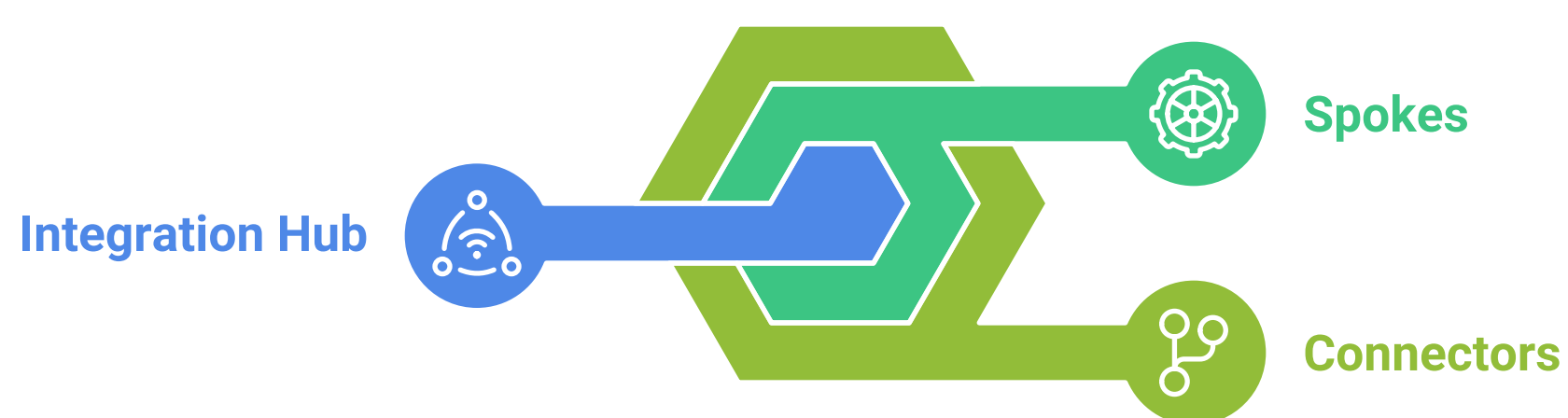
Integration Methods



2. Integration Architecture in ServiceNow

ServiceNow Integration Hub Overview

The Integration Hub in ServiceNow enables integration with both cloud and on-premise systems. The platform provides **Spokes** that contain pre-built actions for common integration scenarios, allowing quick deployment of integrations without needing to code custom solutions. **Connectors** can be configured for multiple third-party systems to extend the platform's capabilities.

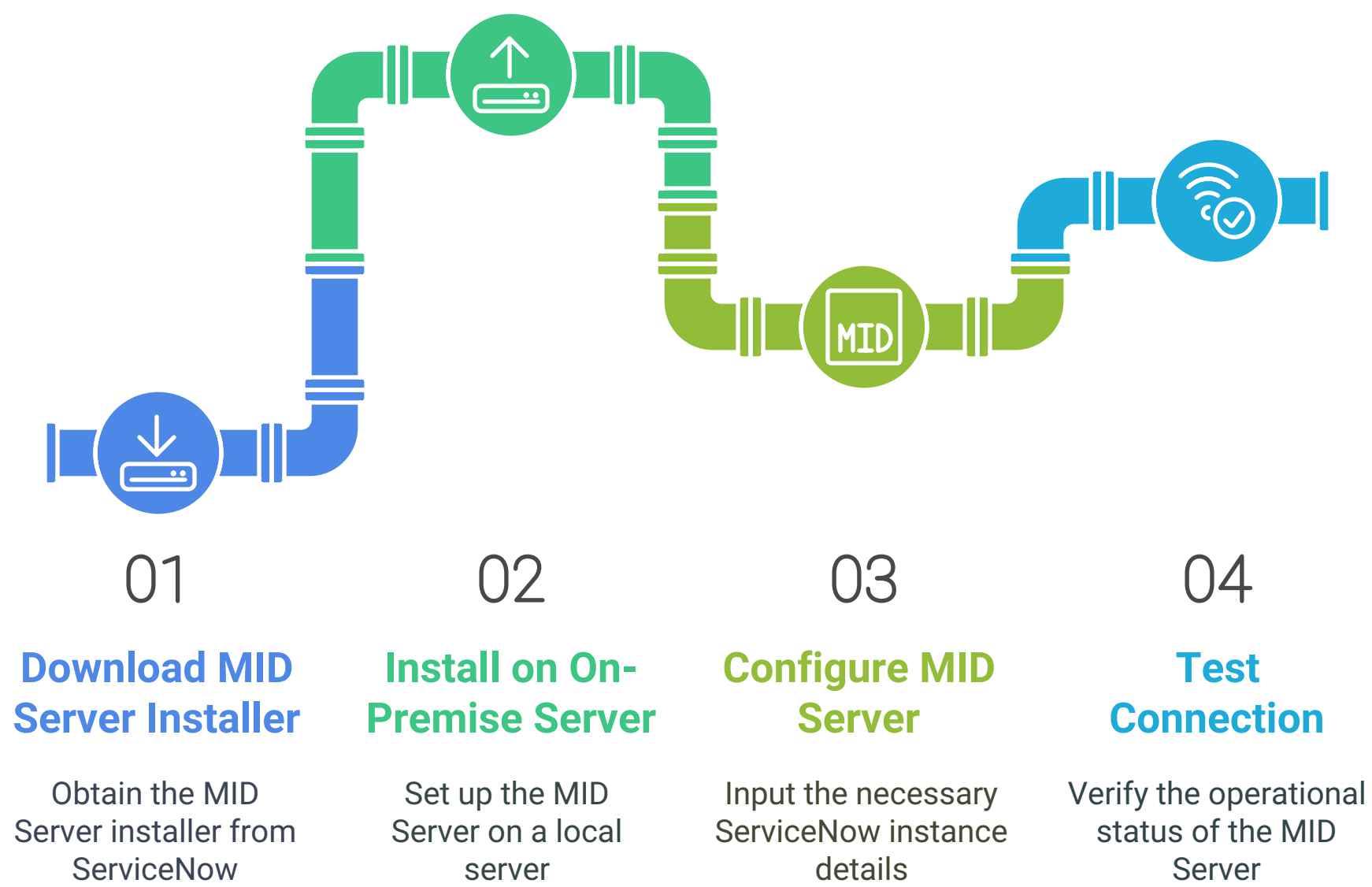


MID Server

The **MID Server** is used to establish secure communication between ServiceNow and on-premise systems or systems behind a firewall. It allows ServiceNow to access data and services on local networks. To install the MID Server, follow these steps:

1. Download the MID Server installer from ServiceNow.
2. Install it on an on-premise server.
3. Configure the MID Server with the correct ServiceNow instance information.
4. Test the connection to ensure it's operational.

MID Server Installation and Configuration



APIs (REST/SOAP)

ServiceNow offers **REST** and **SOAP** web services for integrating with external applications:

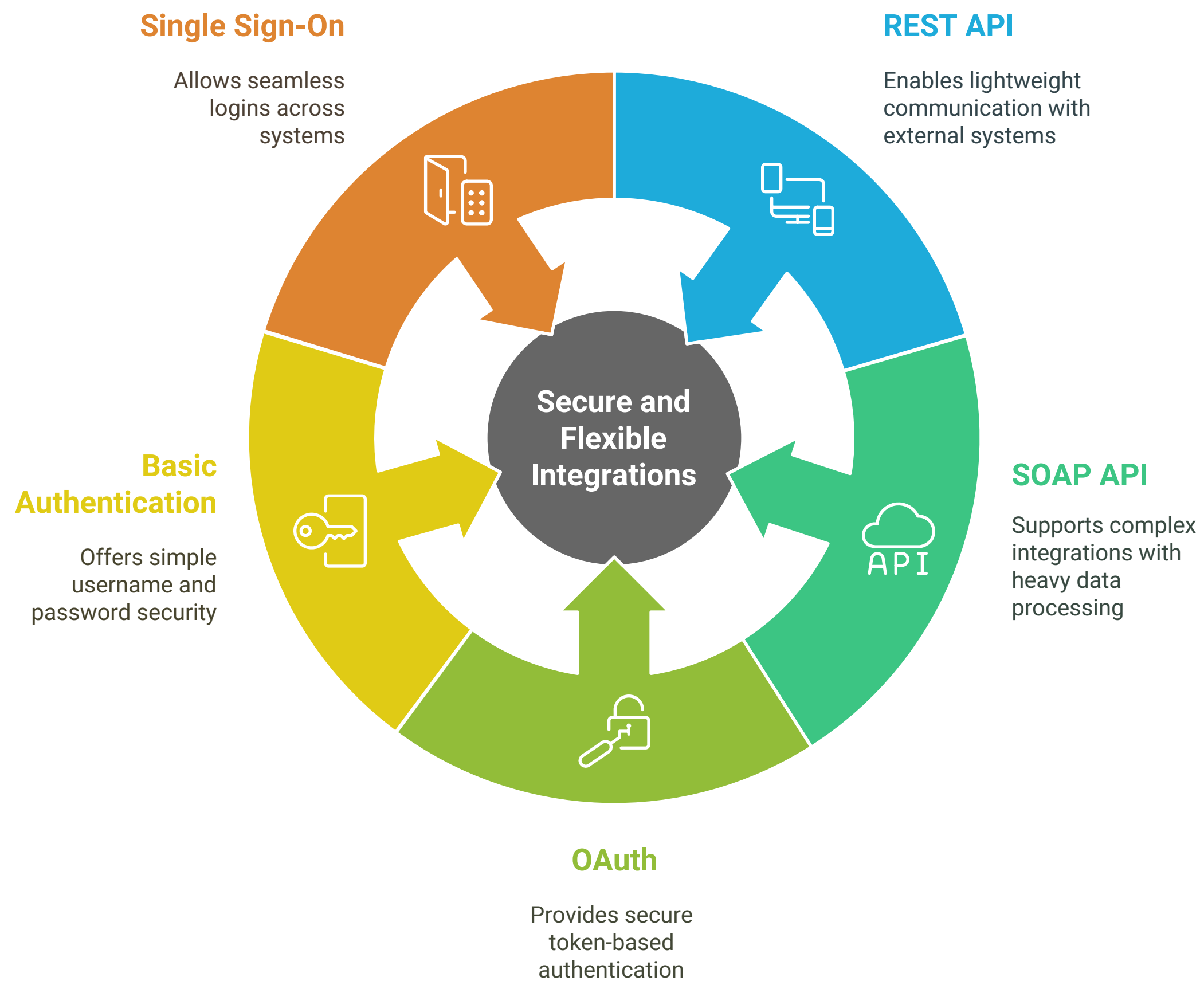
- **REST API:** ServiceNow's REST APIs provide lightweight, flexible communication with external systems.
- **SOAP API:** ServiceNow supports SOAP-based web services, which are ideal for complex integrations requiring heavy data processing or secure communication.

Authentication & Security

To ensure security during integrations:

- **OAuth:** OAuth tokens are used for secure authentication.
- **Basic Authentication:** Used for simpler authentication scenarios.
- **Single Sign-On (SSO):** Facilitates secure logins across integrated systems.

Integration and Security in ServiceNow



3. Integration Methods

Integration Hub

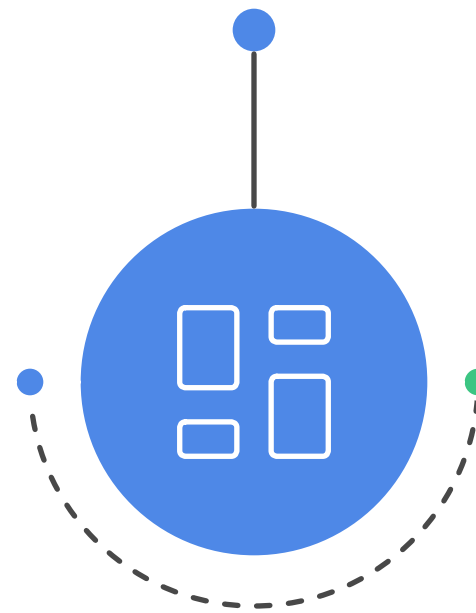
The **Integration Hub** provides a low-code platform for integrating ServiceNow with other tools. It uses **Spokes** to handle various integrations like ServiceNow-to-Cloud, ServiceNow-to-SaaS applications, and more. To use it:

1. Navigate to **Integration Hub** in the ServiceNow dashboard.
2. Choose or create a **Spoke**.
3. Configure **Actions** within the Spoke to define integration tasks [e.g., sending data to a third-party tool].

ServiceNow Integration Hub Process

Navigate to Integration Hub

Access the Integration Hub from the dashboard



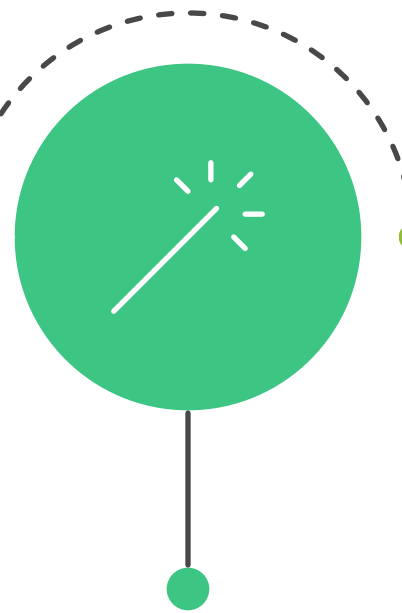
Configure Actions within the Spoke

Define integration tasks by configuring Actions



Choose or create a Spoke

Select or set up a Spoke for integration

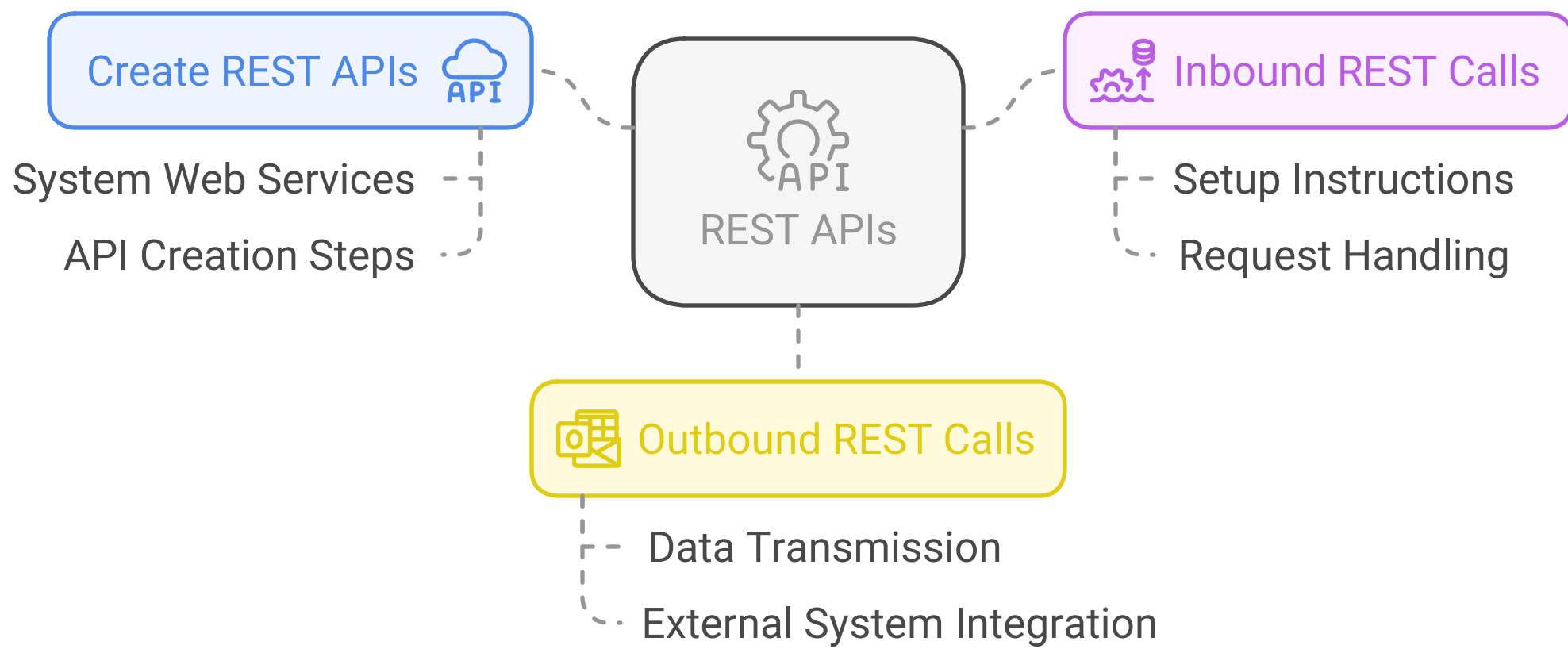


REST APIs

REST APIs enable seamless integration with external systems. Here's how you can configure them:

- **Create REST APIs:** Go to **System Web Services > REST** to create a new API.
- **Inbound REST Calls:** Set up an **Inbound REST API** to handle incoming requests from other systems.
- **Outbound REST Calls:** Configure **Outbound REST** to send data from ServiceNow to external systems.

Configuring REST APIs in ServiceNow



SOAP Web Services

SOAP APIs can be used for integrating with legacy or enterprise systems. The setup involves:

1. Navigate to **System Web Services > SOAP**.
2. Define a new **SOAP Message**.
3. Configure **Inbound/Outbound SOAP** operations.

Setting Up SOAP APIs in ServiceNow



Navigate to SOAP Services

Access the SOAP services menu in ServiceNow



Define SOAP Message

Create a new SOAP message for integration



Configure SOAP Operations

Set up inbound and outbound operations

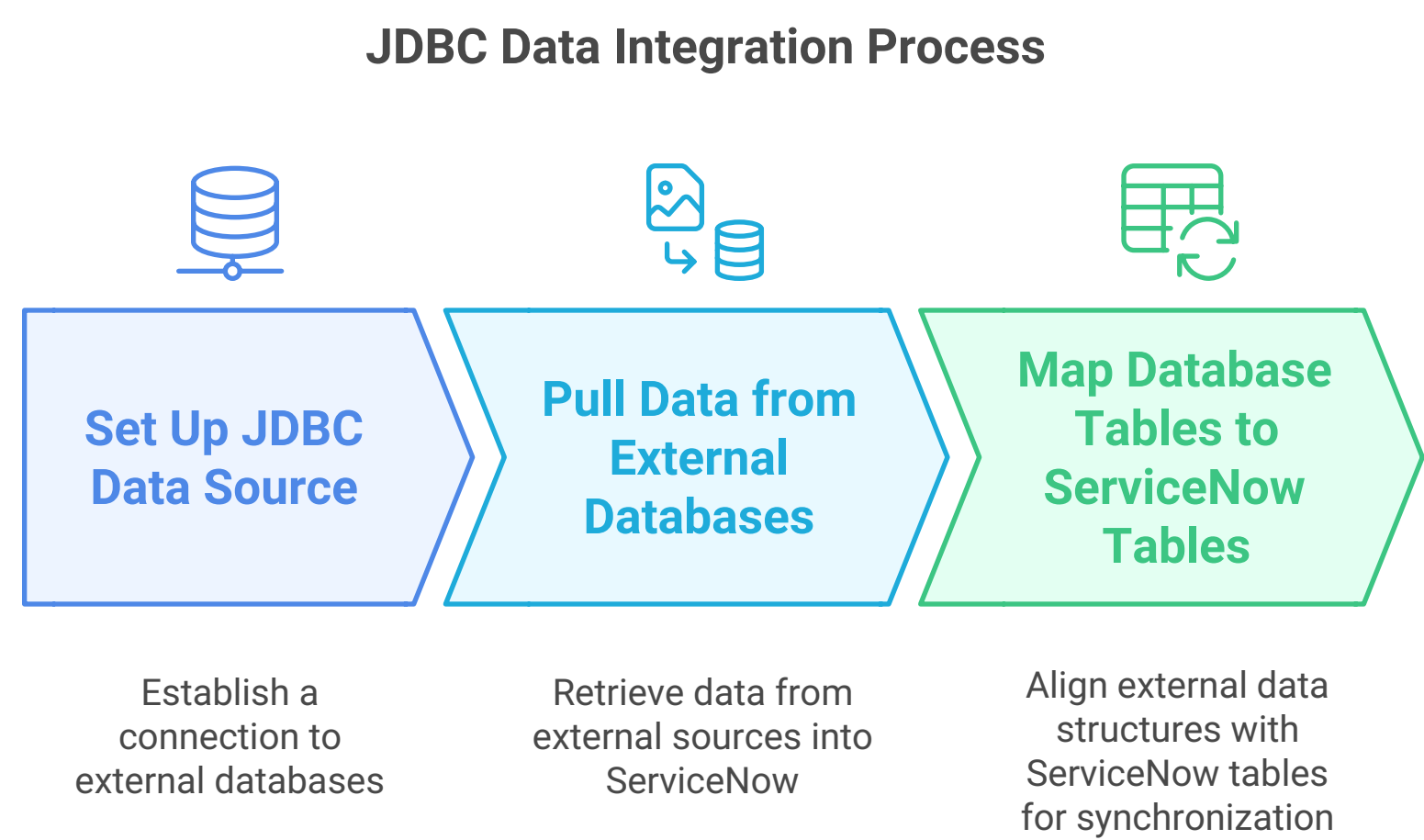


Database Integration

ServiceNow can also integrate with external databases via JDBC or ODBC. Here's how:

- Set up a **JDBC Data Source** within ServiceNow to pull data from external databases.

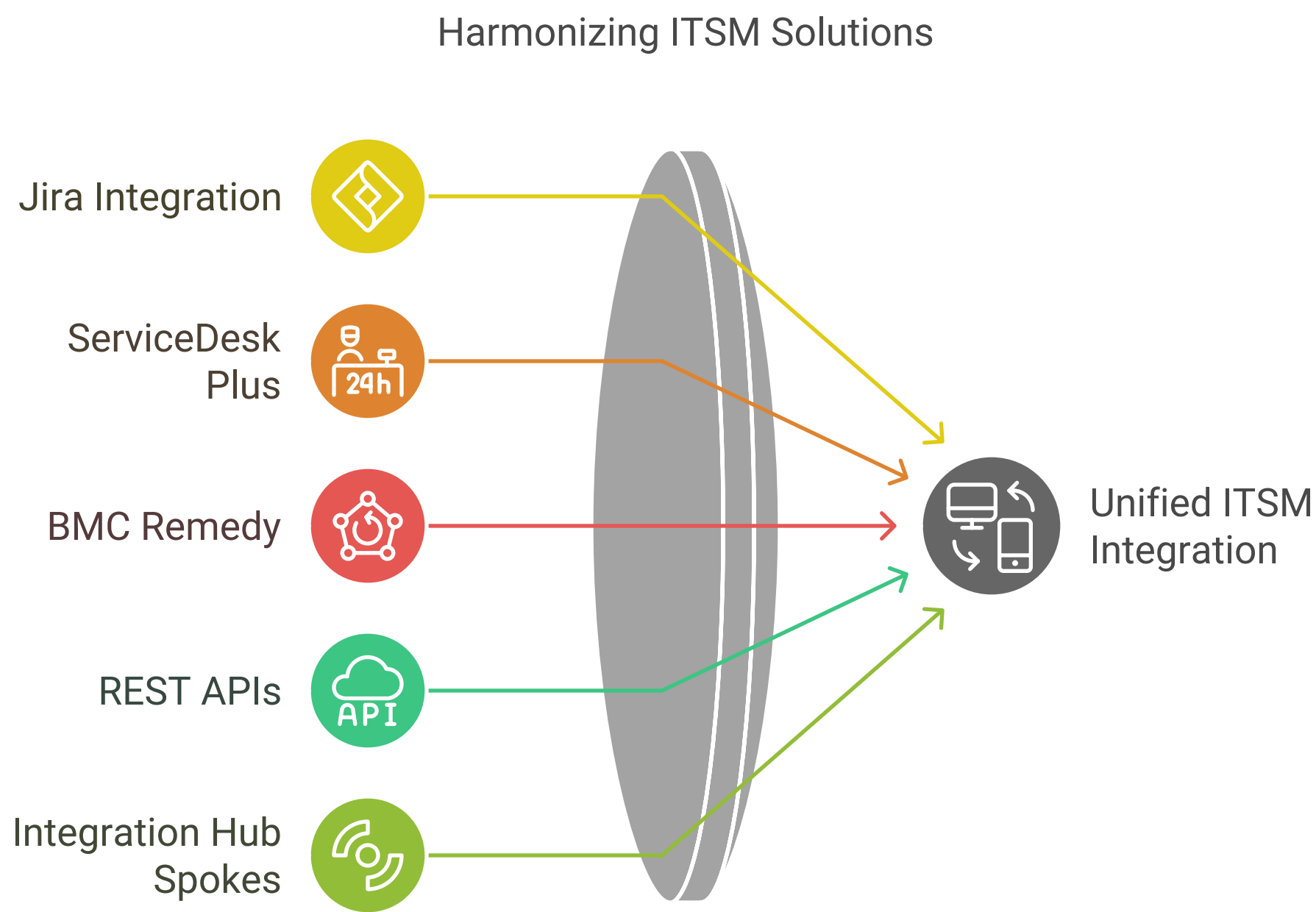
- Map database tables to **ServiceNow tables** to keep the data synchronized.



4. Integration with Common Systems

ITSM Integration

ServiceNow can integrate with other ITSM tools like **Jira**, **ServiceDesk Plus**, or **BMC Remedy**. This integration typically involves mapping incidents, changes, and problems across systems, often using REST APIs or Integration Hub spokes.

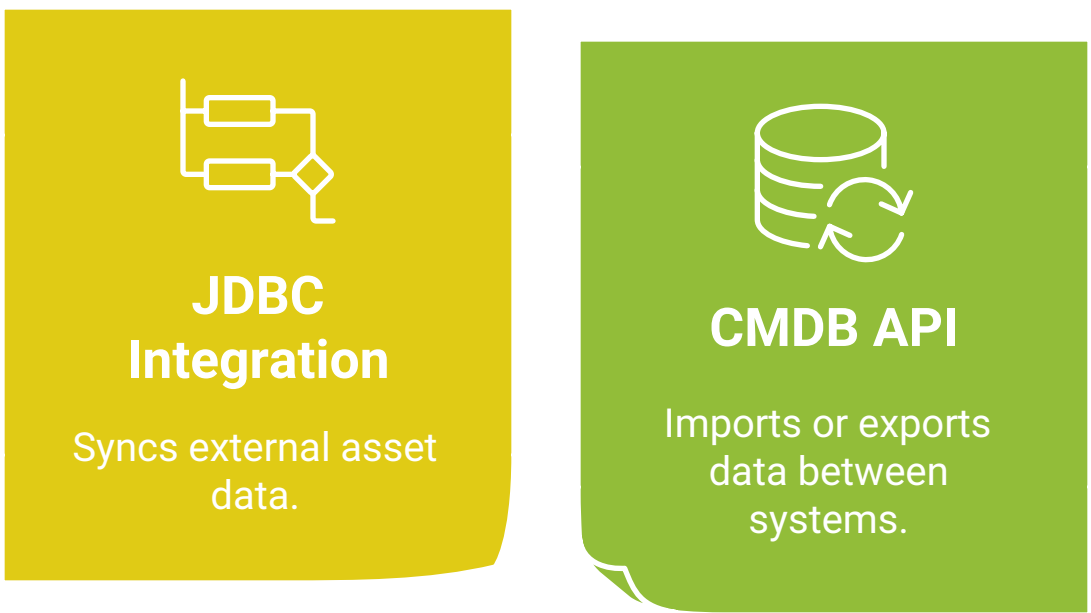


CMDB Integration

Integrating ServiceNow’s **CMDB** with other configuration management systems helps maintain a single source of truth for all IT assets. This can be achieved using:

- **JDBC integration** to sync external asset data.
- **CMDB API** to import or export data between systems.

Data Integration Components

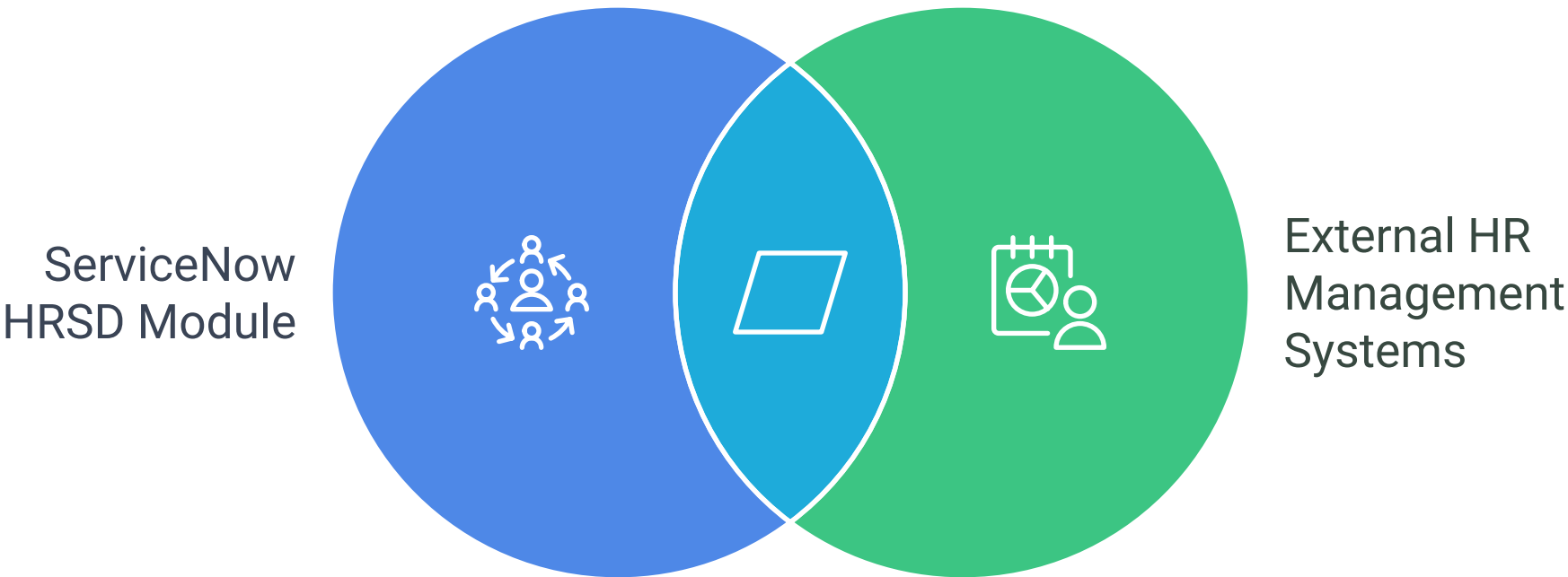


HRSD Integration

ServiceNow’s **HRSD** module can be integrated with HR management systems [like Workday, SAP SuccessFactors, etc.] using REST APIs or Integration Hub spokes. This allows seamless flow of HR data across platforms.

Seamless HR Data Integration

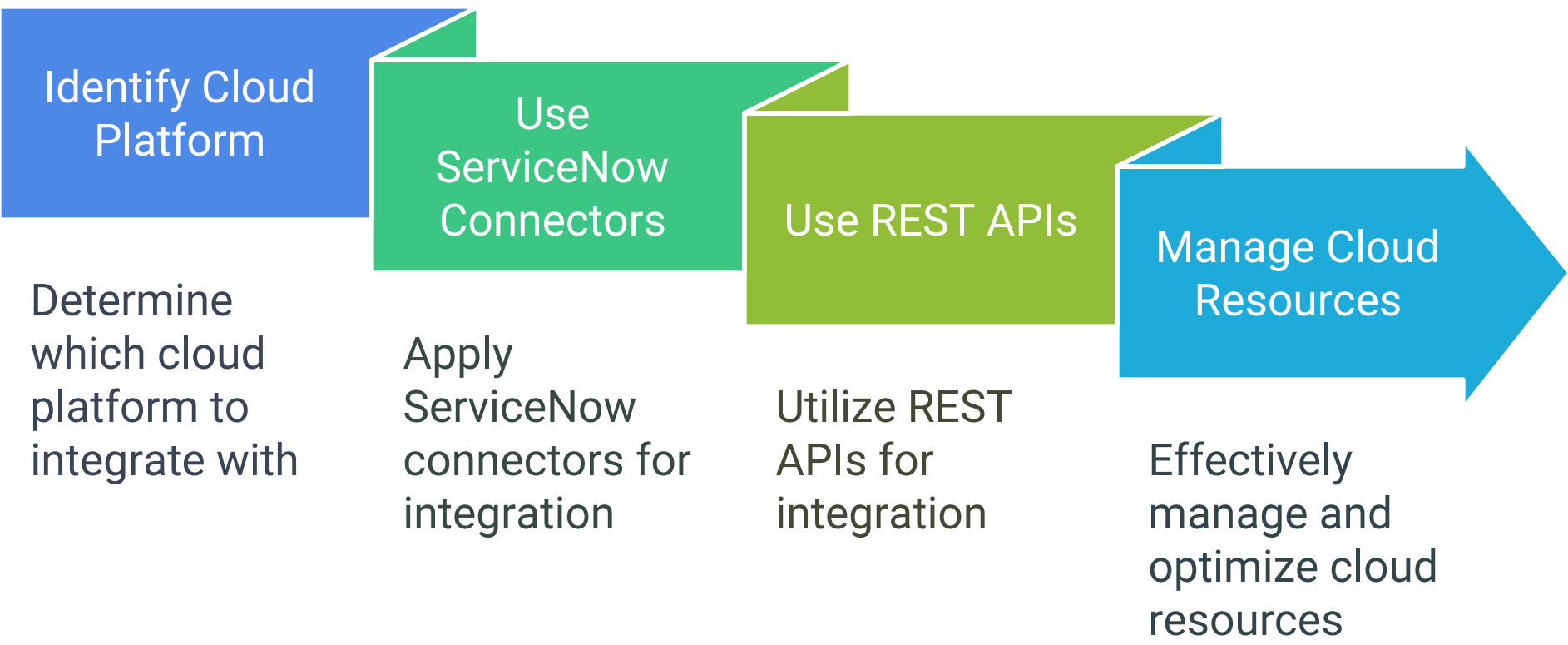
REST APIs and Integration Hub Spokes



Cloud Systems Integration

Integrating with cloud platforms like **AWS**, **Azure**, or **Google Cloud** allows for better management of cloud resources. This integration can be done using ServiceNow connectors or via REST APIs.

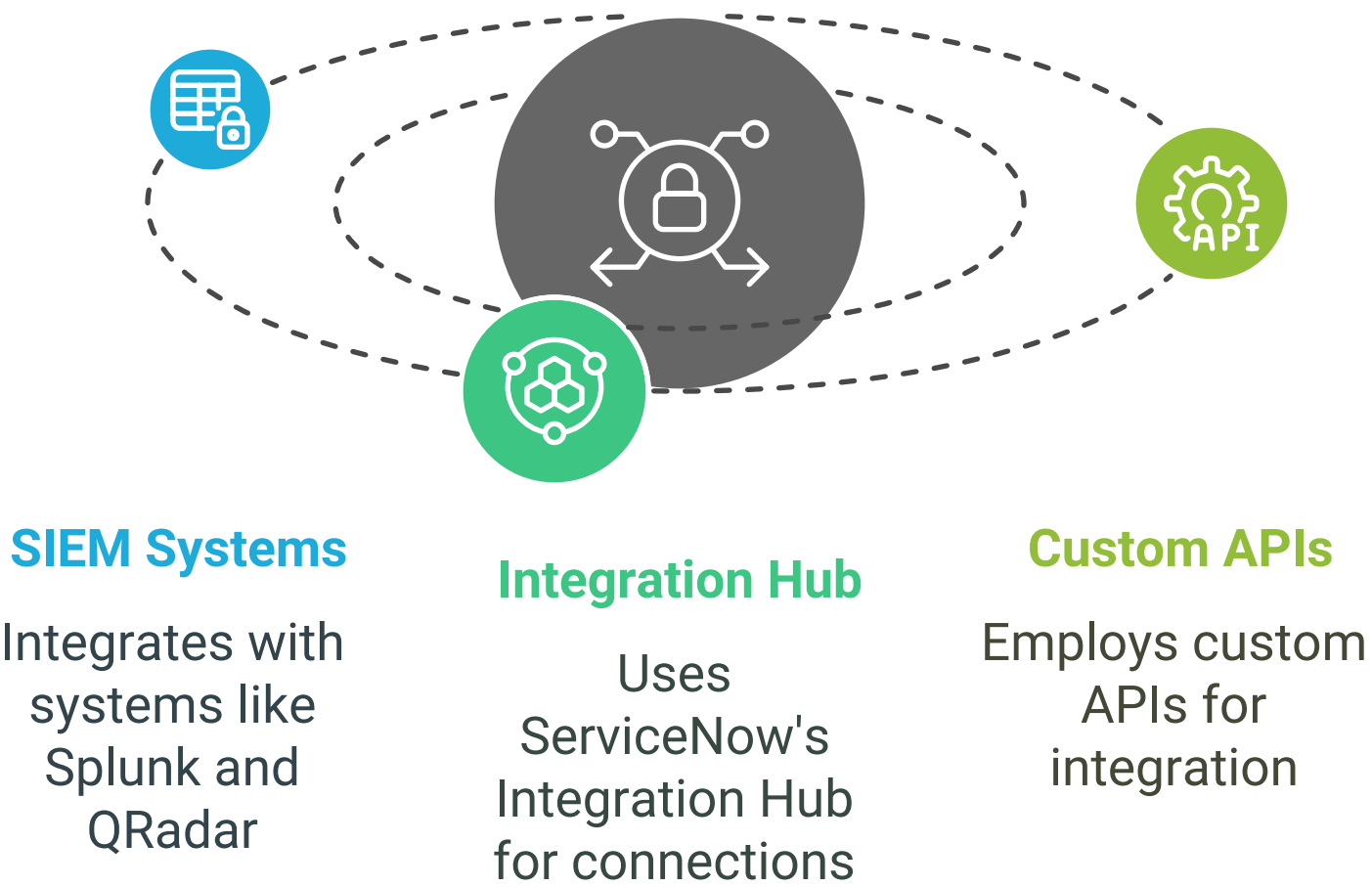
Cloud Integration Process with ServiceNow



Security Operations [SecOps] Integration

ServiceNow's **SecOps** module integrates with security information and event management (SIEM) systems (like Splunk, QRadar) to automate security incidents and workflows. The integration typically involves using the **Integration Hub** or custom APIs.

ServiceNow SecOps Integration Overview



5. Use Case Examples

Incident Management Integration

Scenario: Integrating ServiceNow Incident Management with Jira.

1. Set up a **Jira Spoke** in Integration Hub.
2. Map **Incident fields** to Jira ticket fields.
3. Configure **bi-directional updates** between ServiceNow and Jira for ticket status.

ServiceNow and Jira Integration Process

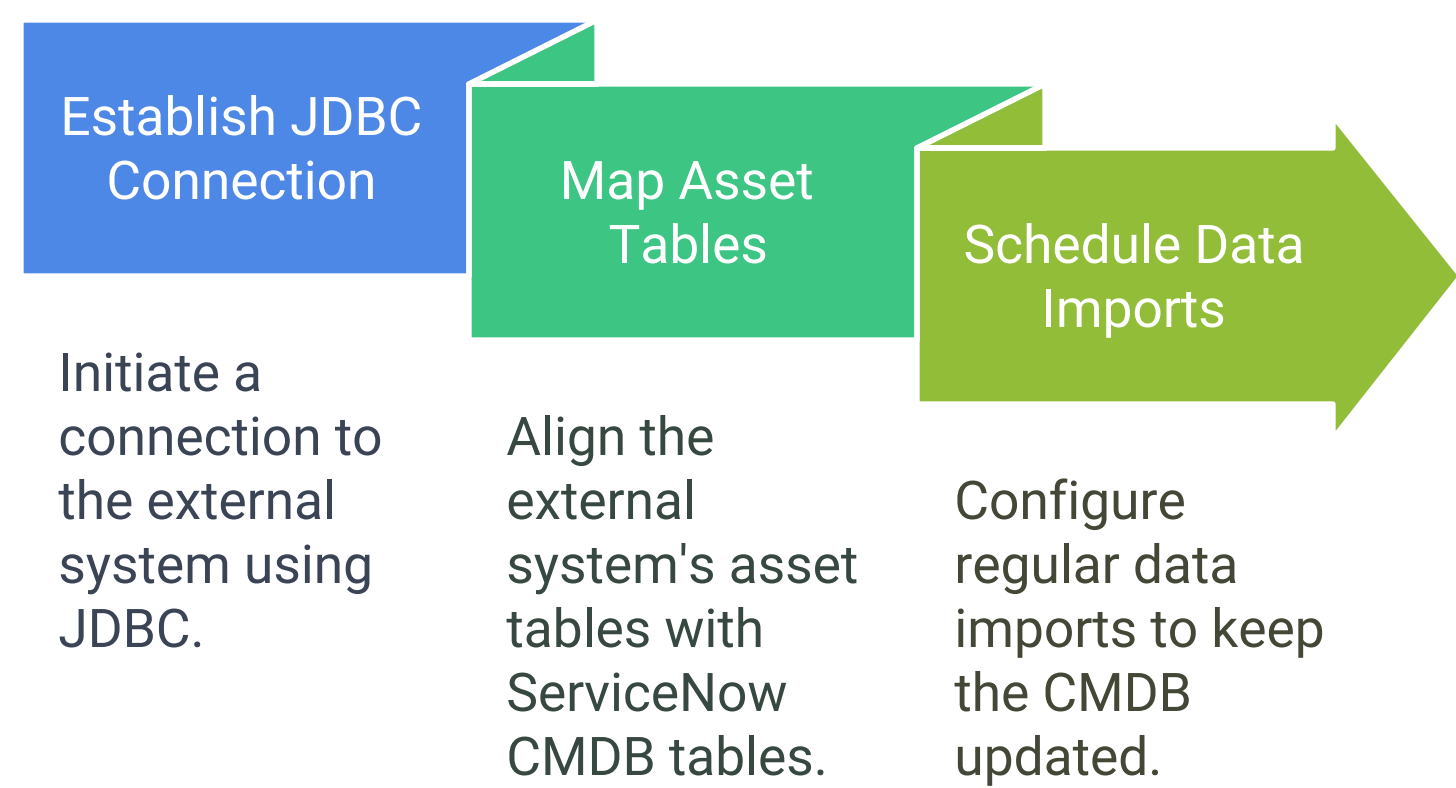


CMDB Synchronization

Scenario: Syncing data between ServiceNow CMDB and an external configuration management system.

1. Use **JDBC** to establish a connection to the external system.
2. Map the external system's **asset tables** to ServiceNow CMDB tables.
3. Set up scheduled data imports to keep CMDB up-to-date.

CMDB Synchronization Process

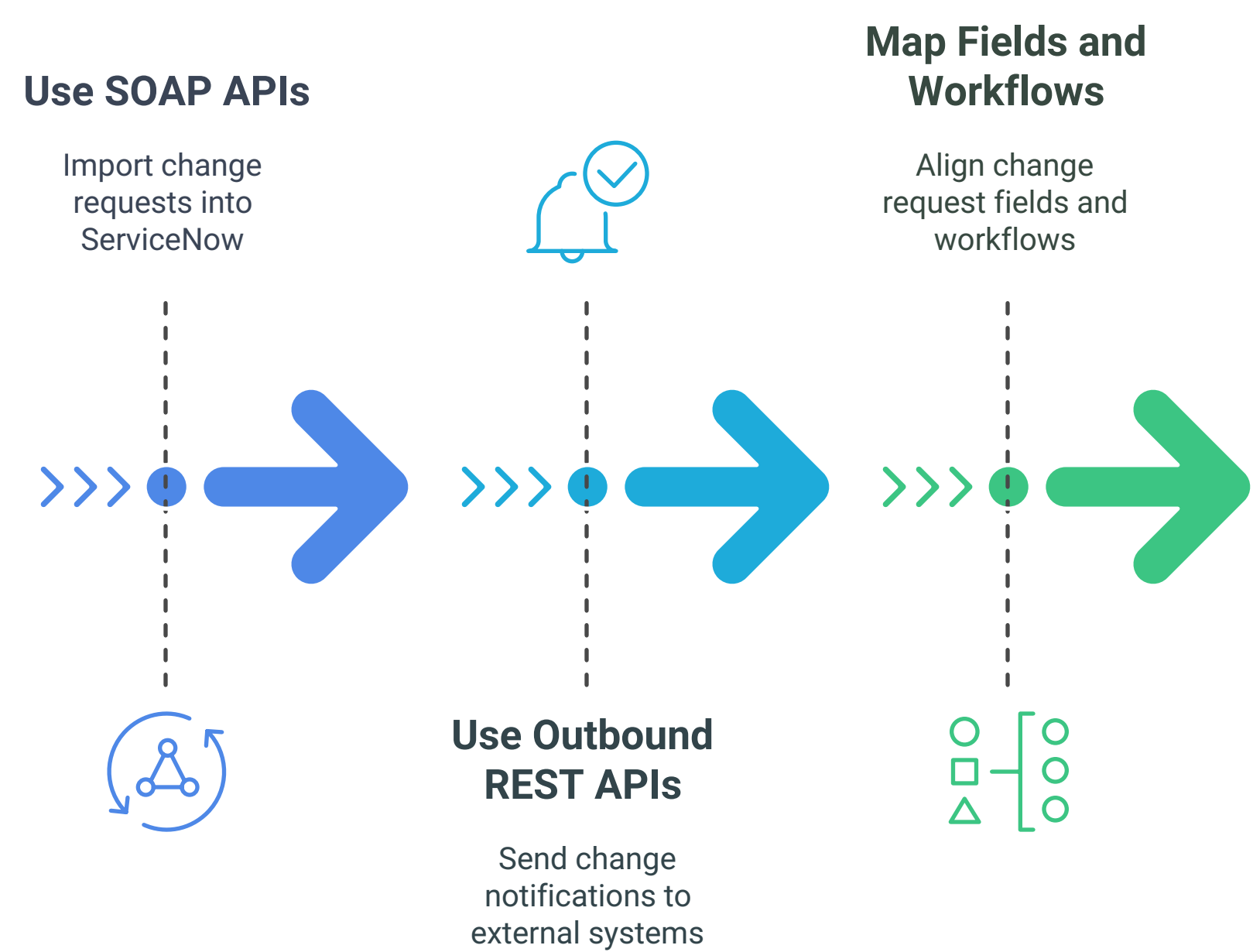


Change Management Workflow

Scenario: Integrating an external change management system with ServiceNow.

- 1. Use **SOAP APIs** to import change requests into ServiceNow.
- 2. Use **Outbound REST APIs** to send change notifications to external systems.
- 3. Map change request fields and workflows accordingly.

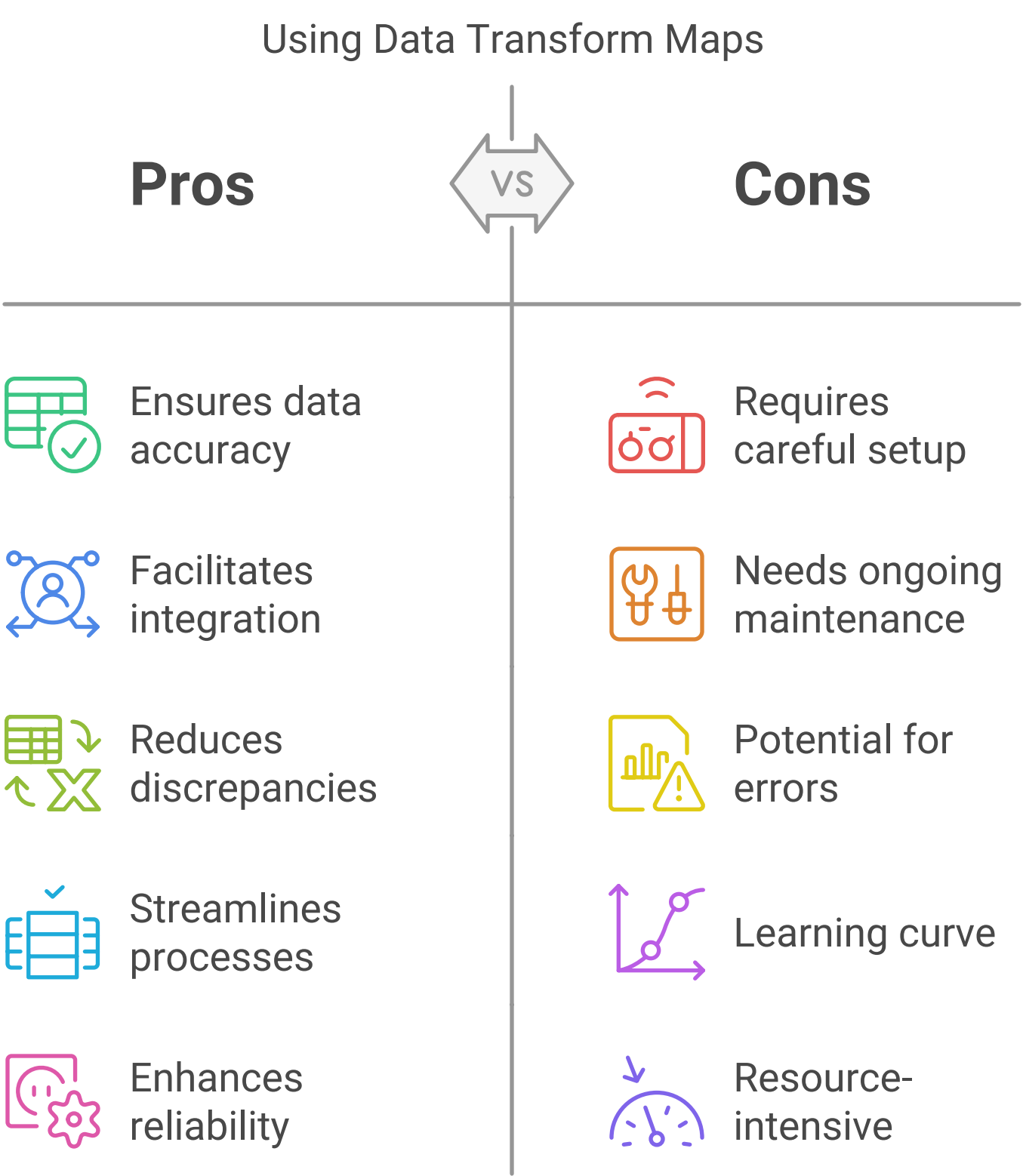
Change Management Integration Sequence



6. Best Practices

Data Mapping and Transformation

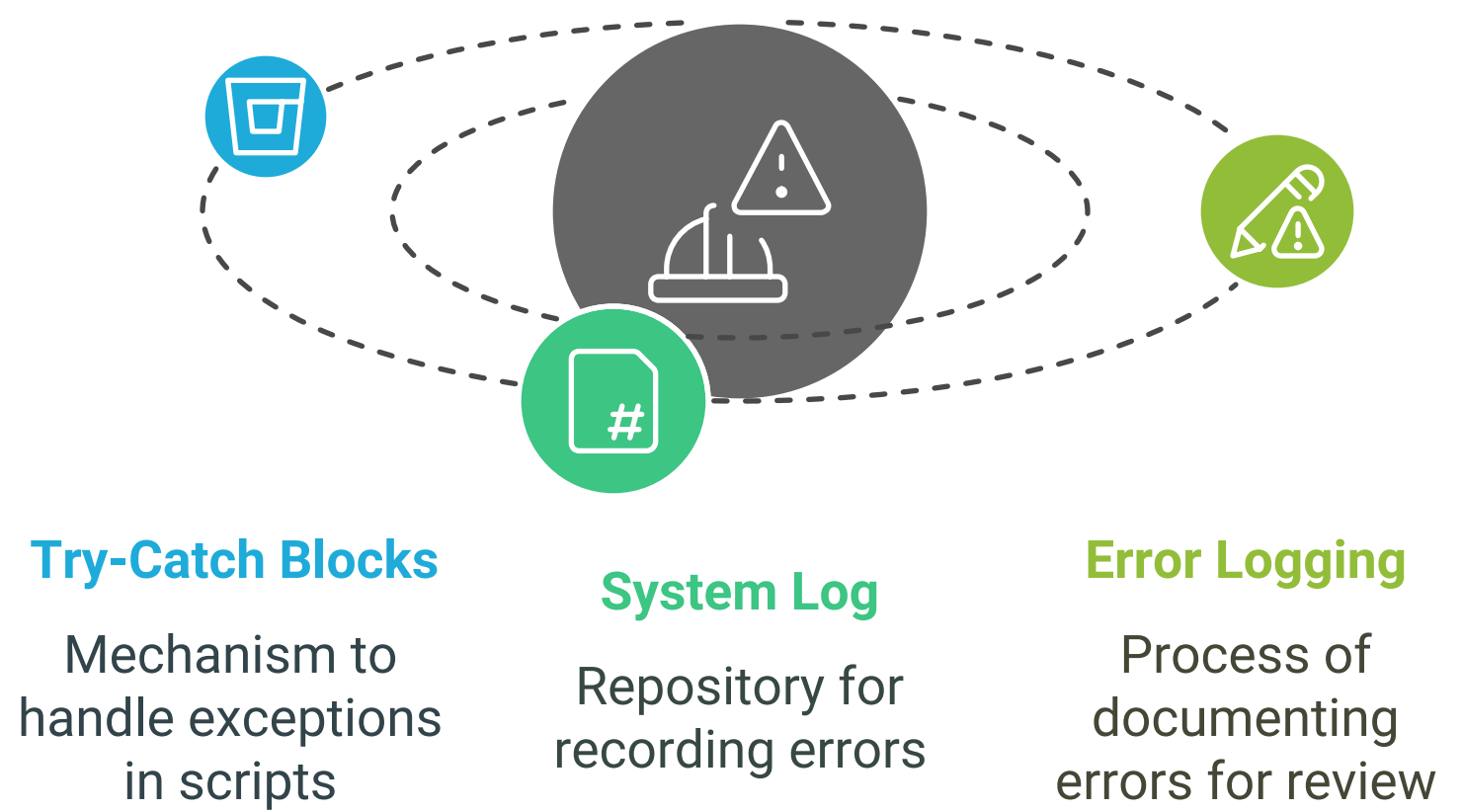
Always ensure that data mapping is done carefully to avoid discrepancies. Use **Data Transform Maps** to map data between ServiceNow tables and external systems.



Error Handling and Logging

Ensure proper **error handling** by using try-catch blocks in scripts and logging errors to the **System Log**.

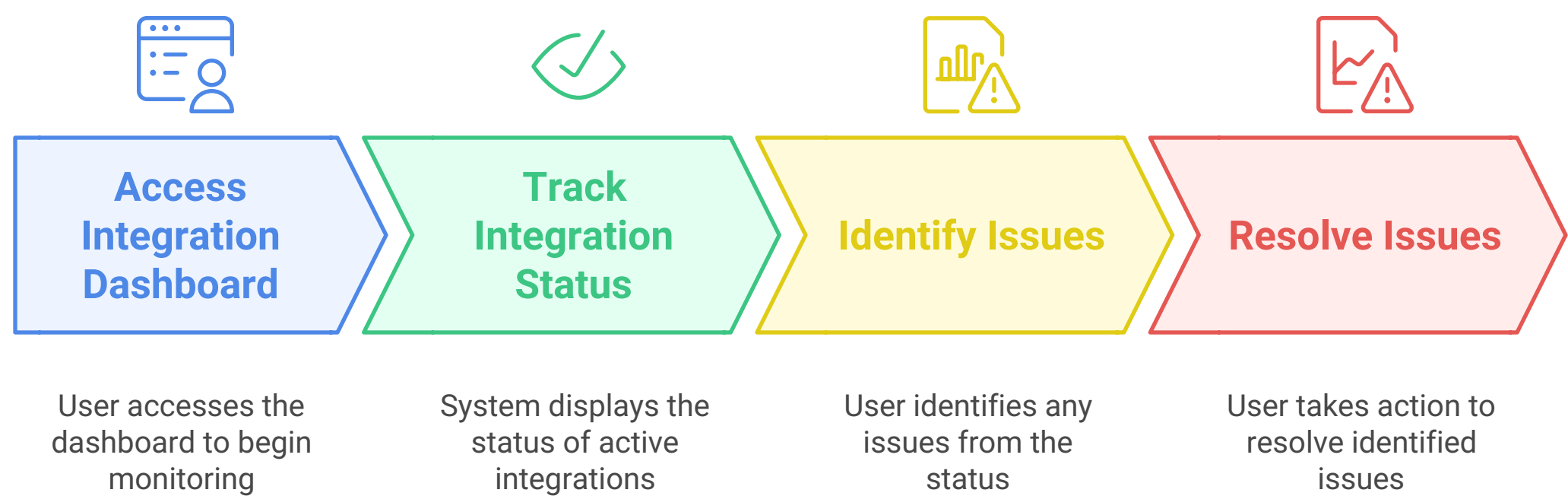
Effective Error Handling in ServiceNow



Monitoring Integrations

Monitor integrations through the **Integration Dashboard**. This allows you to track the status of active integrations and resolve issues quickly.

Monitoring Integrations Sequence



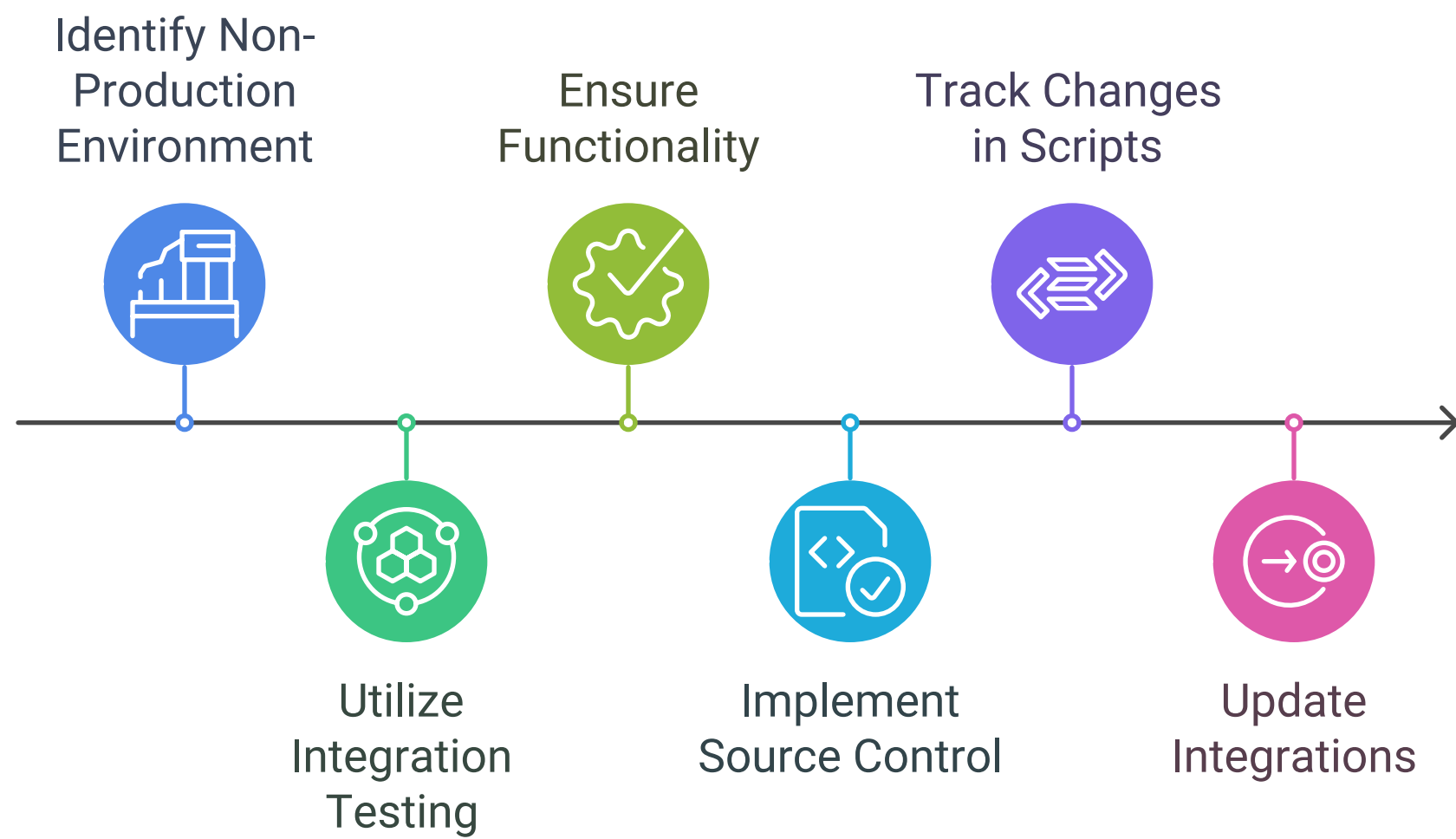
Testing

Before deployment, test integrations in a **non-production environment**. ServiceNow provides **Integration Testing** capabilities to ensure everything is functioning properly.

Version Control

Use **Source Control** in ServiceNow to track changes in integration scripts and ensure smooth updates to existing integrations.

Integration Testing and Version Control in ServiceNow

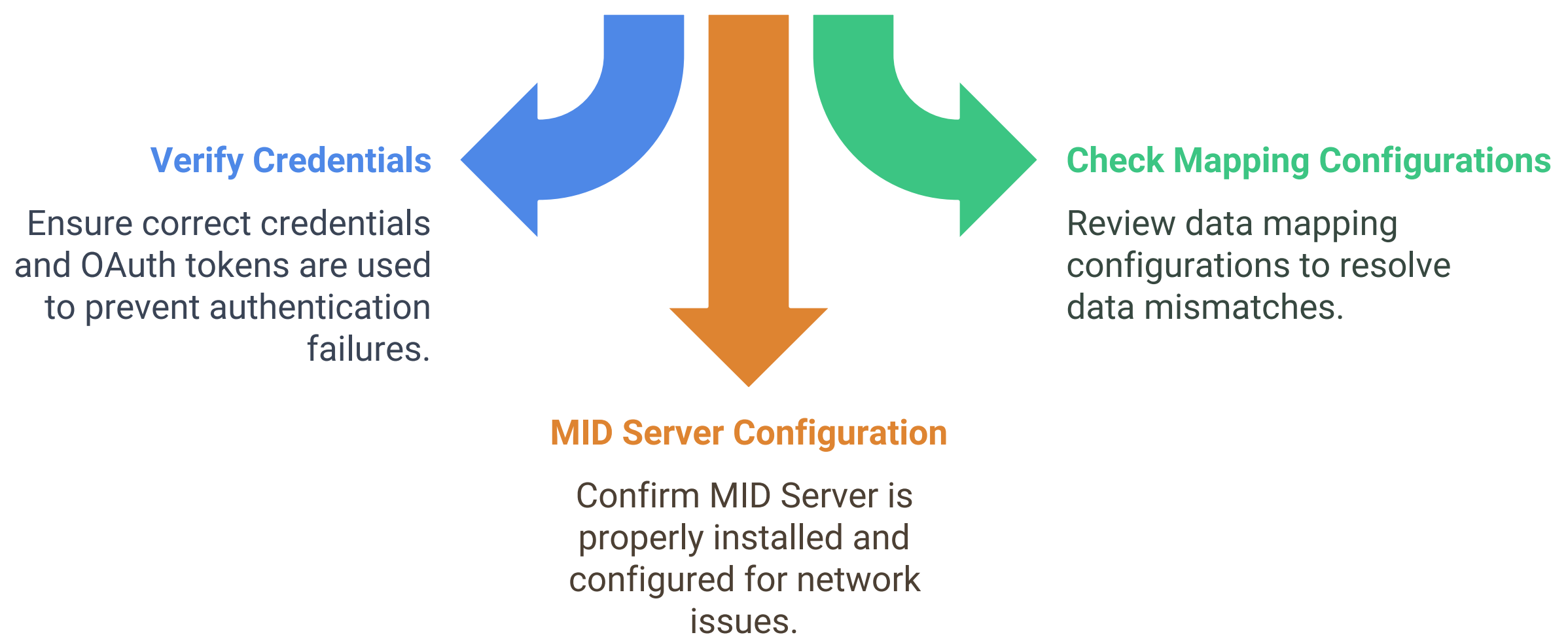


7. Troubleshooting

Common Integration Issues

- **Authentication failures:** Verify credentials and OAuth tokens.
- **Data mismatches:** Check mapping configurations.
- **Network issues:** Ensure the MID Server is installed and properly configured for communication.

How to troubleshoot integration issues?



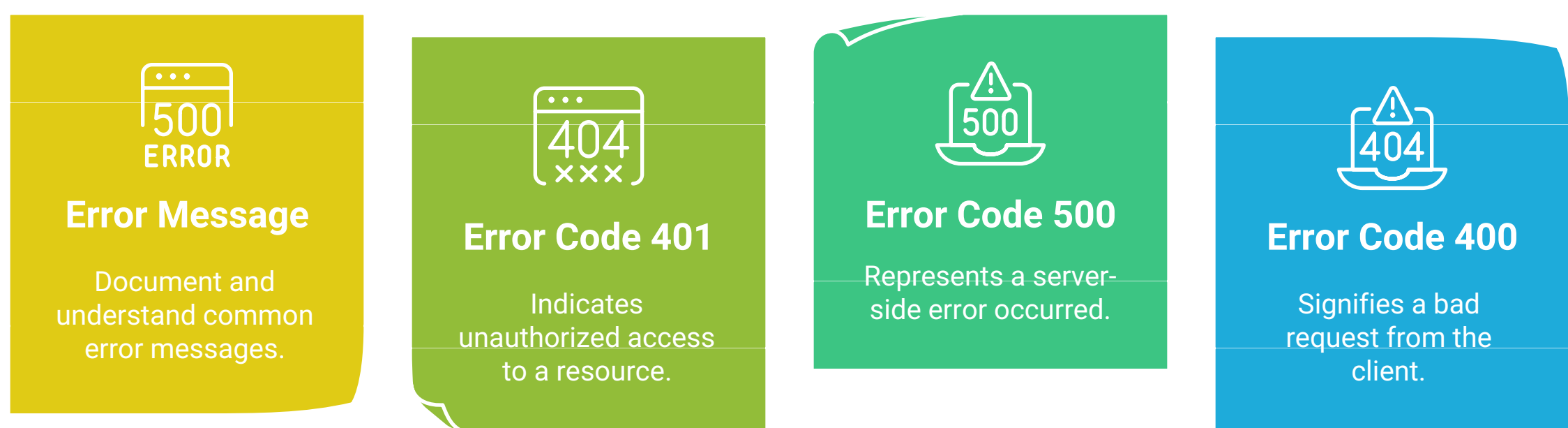
Logs and Debugging Tools

Use the **System Logs** and **MID Server logs** for detailed troubleshooting. Additionally, use the **REST API Explorer** to test API calls and validate responses.

Error Messages

Ensure you document and understand the common error messages like **401 Unauthorized**, **500 Internal Server Error**, and **400 Bad Request**.

Common Error Messages



Regularly monitor integration logs and performance metrics. Update integrations as systems evolve or when updates are released for ServiceNow.

Scaling Integrations

As your ServiceNow instance grows, scale integrations by optimizing scripts, using queuing systems for batch processes, and ensuring that APIs are rate-limited.

Security Updates

Apply security patches and update integration scripts regularly to keep them secure.

ServiceNow Integration Maintenance Cycle



9. Advanced Topics

Complex Integrations

Complex integrations involve multiple systems, custom scripts, and advanced authentication techniques. In these cases, the integration process may involve:

- **Custom Web Services**
- **Asynchronous Data Processing**
- **B2B integrations**



Custom Web Services

Services tailored for specific integration needs.



Asynchronous Data Processing

Handling data in a non-blocking manner.



B2B Integrations

Connecting business systems for seamless communication.